

Healthcare Insurance Cost Analysis

SQL | Power BI | Healthcare Analytics | Financial Analysis | Business Intelligence | Reporting

Portfolio Project

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About This Project

Healthcare insurance providers generate vast amounts of customer, policy, and financial data every day. Transforming this data into actionable business intelligence is essential for improving underwriting decisions, managing financial risk, optimising premium pricing, and ensuring long-term organisational profitability.

This project presents an end-to-end **Healthcare Insurance Cost Analysis** solution developed using **Microsoft SQL Server** and **Microsoft Power BI**. The objective was to analyse healthcare insurance data, identify the key factors influencing insurance charges, and translate analytical findings into practical business insights for executive decision-making.

The project follows a complete business intelligence workflow—from SQL data extraction and preparation to interactive dashboard development and executive reporting. By examining variables such as **age, gender, Body Mass Index (BMI), smoking status, number of children, and geographic region**, the analysis identifies the primary drivers of healthcare expenditure and evaluates their impact on insurance pricing.

Rather than focusing solely on technical implementation, this project demonstrates how modern data analytics supports strategic business objectives by enabling insurers to make evidence-based decisions, improve pricing accuracy, strengthen risk management, enhance financial planning, and promote preventive healthcare initiatives.

Designed from the perspective of a Business Intelligence Analyst, Financial Analyst, and Business Analyst, the project highlights the value of combining technical expertise with business understanding to solve real-world organisational challenges.

This portfolio project showcases practical capabilities in **SQL, Microsoft Power BI, healthcare analytics, business intelligence, financial analysis, KPI reporting, executive dashboard development, and data storytelling**—skills that are highly relevant to analytical roles within multinational organisations such as **Stripe, Microsoft, Amazon, EY, Deloitte, Accenture, AIB, Bank of Ireland, HubSpot**, and other global, data-driven enterprises.

Executive Summary

Healthcare organisations and insurance providers operate in an environment where accurate pricing decisions, effective risk assessment, and sustainable financial performance depend heavily on data-driven insights. As healthcare costs continue to rise globally, insurers require analytical solutions that identify the underlying factors influencing medical expenses while supporting fair, evidence-based premium pricing.

This project presents a comprehensive Healthcare Insurance Cost Analysis developed using **Microsoft SQL Server** and **Microsoft Power BI**. The analysis transforms raw insurance policy data into interactive business intelligence dashboards that enable decision-makers to understand the primary drivers of healthcare costs across different customer segments.

Through structured SQL data extraction, data modelling, interactive visualisations, and executive reporting, the project evaluates how demographic characteristics and lifestyle factors—including age, body mass index (BMI), smoking status, gender, family size, and geographic region—influence insurance charges.

The findings demonstrate that behavioural and health-related factors have a substantially greater impact on insurance costs than demographic variables alone. These insights support more accurate underwriting decisions, improved pricing strategies, stronger financial planning, and enhanced long-term profitability.

Beyond technical implementation, this project illustrates how modern data analytics can help insurance organisations transform operational data into strategic business intelligence that supports executive decision-making.

Business Context

Health insurance companies manage significant financial risk by estimating future healthcare costs and determining appropriate premium pricing for millions of policyholders.

Accurate pricing is essential because premiums collected today must be sufficient to cover future medical claims while maintaining sustainable profitability and regulatory compliance.

Historically, many pricing decisions relied heavily on historical averages and actuarial judgement. However, advances in data analytics now enable insurers to analyse customer behaviour, health characteristics, and historical claims patterns with significantly greater precision.

This project demonstrates how SQL and Power BI can be applied to analyse healthcare insurance data and identify the key variables driving insurance expenditure.

The analytical framework provides management with actionable insights into customer risk profiles, allowing organisations to optimise pricing models, improve underwriting decisions, reduce financial exposure, and promote preventive healthcare initiatives.

Rather than functioning solely as a reporting exercise, the project illustrates how business intelligence supports strategic decision-making across underwriting, finance, operations, and executive leadership.

Project Objectives

The primary objective of this project was to develop an interactive business intelligence solution capable of analysing healthcare insurance costs and identifying the major factors influencing premium pricing.

Specific objectives included:

- Extract insurance data using Microsoft SQL Server.
- Clean and prepare datasets for analytical reporting.
- Develop an interactive Power BI dashboard.
- Analyse relationships between customer characteristics and insurance charges.
- Identify the strongest predictors of healthcare expenditure.
- Evaluate demographic and lifestyle factors affecting insurance costs.
- Provide executive-level insights supporting pricing and underwriting decisions.
- Develop practical recommendations to improve financial performance and operational efficiency.

The project also demonstrates how data analytics can bridge the gap between technical reporting and strategic business decision-making.

Business Questions

The analysis was designed to answer several key business questions commonly faced by insurance providers.

Customer Risk

- Which customer characteristics contribute most significantly to healthcare insurance costs?
- Which policyholders present the highest financial risk?

Pricing Strategy

- Should insurance pricing be influenced more by demographic characteristics or lifestyle behaviours?
- Which variables should receive greater weighting within premium pricing models?

Healthcare Cost Drivers

- How does smoking influence healthcare expenditure?
- What relationship exists between BMI and insurance charges?
- Does age significantly increase healthcare costs?
- How does family size affect insurance expenditure?

Customer Segmentation

- Are there meaningful differences in healthcare costs across gender?
- Do insurance costs vary significantly across geographic regions?

Financial Performance

- Which customer groups contribute most to future claims costs?
- How can insurers improve profitability while maintaining equitable pricing practices?

Answering these questions enables insurance companies to make more informed underwriting decisions while improving long-term financial sustainability.

Technology Stack

The project was developed using widely adopted enterprise analytics technologies commonly utilised within healthcare organisations, insurance companies, financial institutions, and multinational corporations.

Database

- Microsoft SQL Server

Data Extraction

- SQL
- Filtering
- Sorting
- Aggregate Functions

Business Intelligence

- Microsoft Power BI

Data Preparation

- Power Query
- Data Cleaning
- Data Transformation
- Data Modelling

Data Visualisation

- KPI Cards
- Interactive Dashboards
- Charts
- Slicers
- Cross Filtering

Analytics

- Descriptive Analytics
- Customer Segmentation
- Healthcare Cost Analysis
- Business Intelligence Reporting

Methodology

The project followed a structured end-to-end analytics workflow aligned with industry best practices for business intelligence and healthcare analytics.

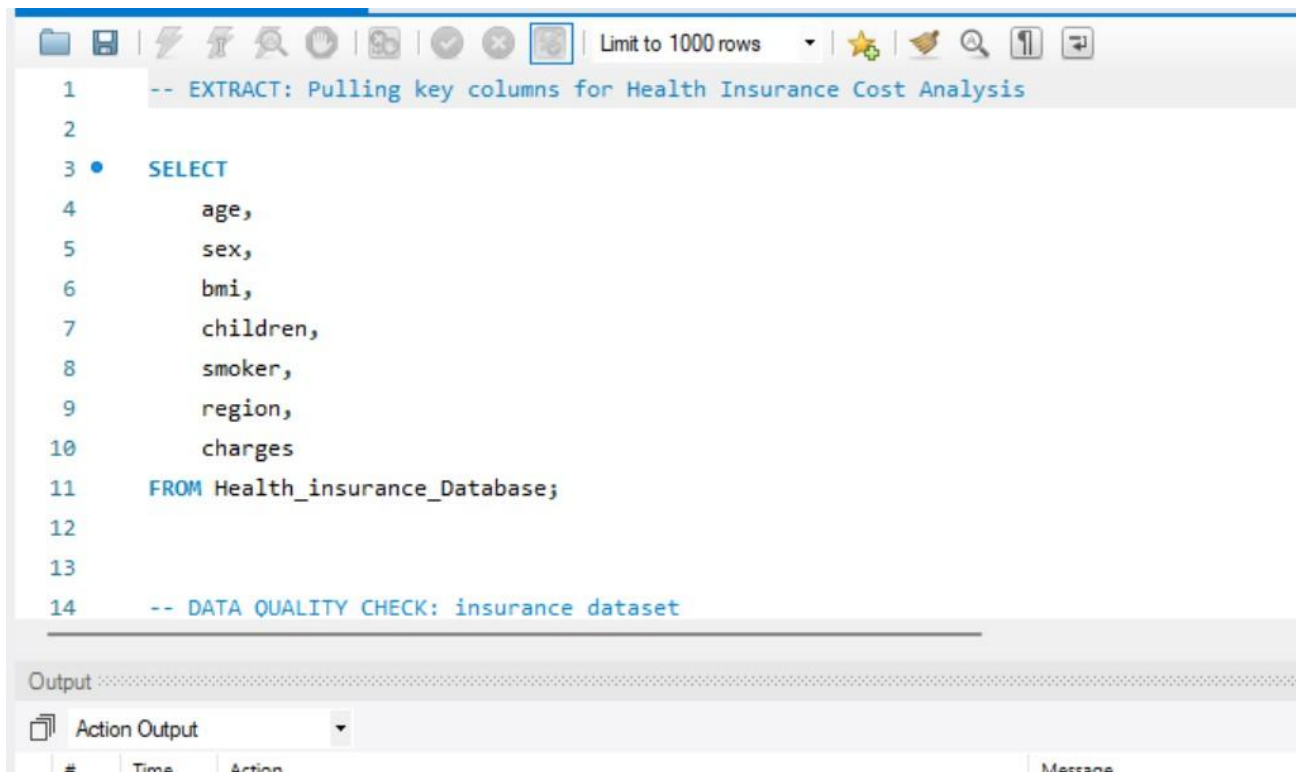
Phase 1 — Data Extraction

Healthcare insurance records were extracted from a structured database using Microsoft SQL Server.

SQL queries were developed to retrieve key policyholder information, including:

- Age
- Gender
- Body Mass Index (BMI)
- Number of Children
- Smoking Status
- Geographic Region
- Insurance Charges

Additional SQL techniques—including filtering, sorting, and aggregation—were applied to prepare a clean analytical dataset suitable for reporting.



The screenshot displays a SQL query editor interface. The query text is as follows:

```
1  -- EXTRACT: Pulling key columns for Health Insurance Cost Analysis
2
3  •  SELECT
4      age,
5      sex,
6      bmi,
7      children,
8      smoker,
9      region,
10     charges
11  FROM Health_insurance_Database;
12
13
14  -- DATA QUALITY CHECK: insurance dataset
```

The interface includes a toolbar at the top with icons for file operations, execution, and search. A status bar at the bottom shows 'Output' and 'Action Output' tabs, along with a 'Messages' pane.

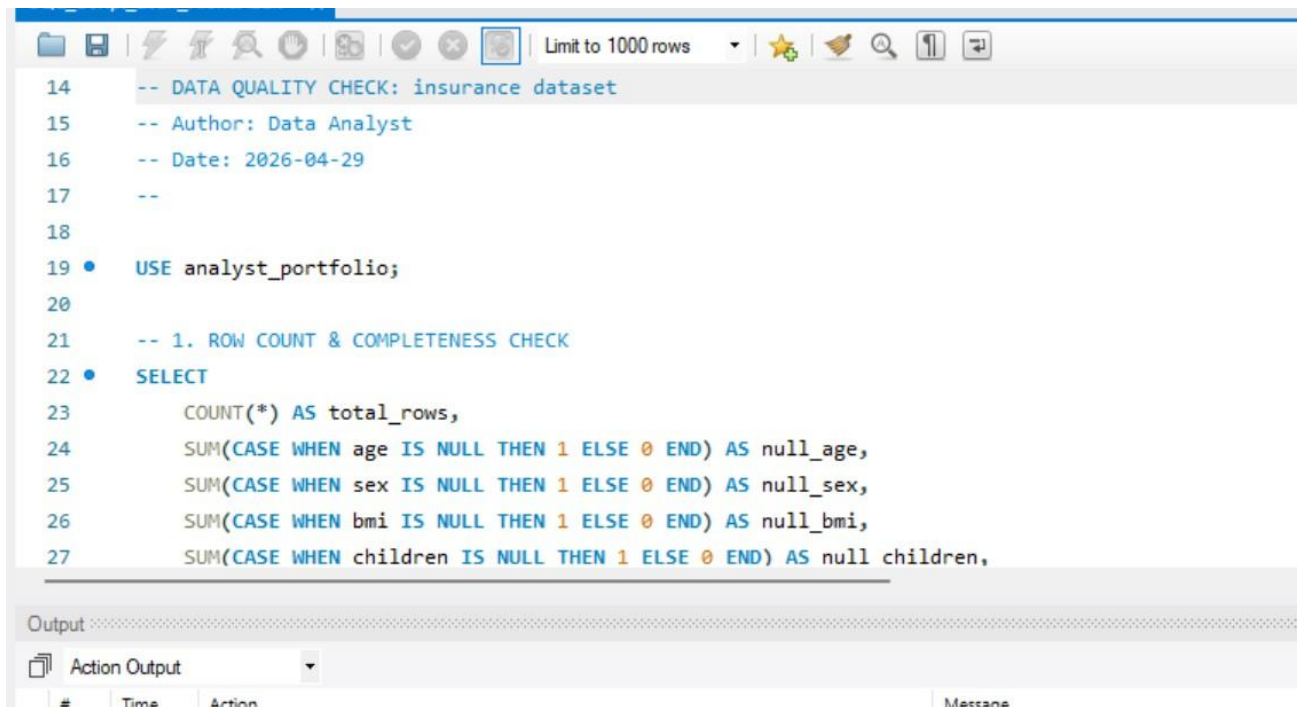
Phase 2 — Data Preparation

The extracted data was imported into Microsoft Power BI, where Power Query was used to perform data cleaning and transformation.

Data preparation activities included:

- Data validation
- Quality checks
- Data modelling
- Relationship management
- Optimisation for reporting

These steps ensured that the dataset was accurate, consistent, and suitable for interactive business intelligence analysis.



```
14  -- DATA QUALITY CHECK: insurance dataset
15  -- Author: Data Analyst
16  -- Date: 2026-04-29
17  --
18
19  •  USE analyst_portfolio;
20
21  -- 1. ROW COUNT & COMPLETENESS CHECK
22  •  SELECT
23      COUNT(*) AS total_rows,
24      SUM(CASE WHEN age IS NULL THEN 1 ELSE 0 END) AS null_age,
25      SUM(CASE WHEN sex IS NULL THEN 1 ELSE 0 END) AS null_sex,
26      SUM(CASE WHEN bmi IS NULL THEN 1 ELSE 0 END) AS null_bmi,
27      SUM(CASE WHEN children IS NULL THEN 1 ELSE 0 END) AS null_children,
```

Output

Action Output

#	Time	Action	Message
---	------	--------	---------

Phase 3 — Dashboard Development

An interactive Power BI dashboard was developed to enable executives and business users to explore insurance costs dynamically.

The dashboard incorporates:

- KPI Cards
- Interactive Filters
- Cross-filtering
- Trend Visualisations
- Customer Segmentation
- Comparative Analysis

The interactive design allows users to analyse insurance charges from multiple business perspectives without requiring technical expertise.

Phase 4 — Business Analysis

Using SQL outputs and Power BI visualisations, the project evaluated the influence of several variables on healthcare insurance costs, including:

- Smoking Status
- Age
- Body Mass Index (BMI)
- Gender
- Number of Children
- Geographic Region

Each variable was assessed individually before being interpreted collectively to understand its contribution to overall healthcare expenditure.

Phase 5 — Executive Reporting

The final stage translated analytical findings into business-focused recommendations suitable for executive decision-makers.

Rather than presenting technical outputs alone, the project communicates insights through clear narratives, KPI reporting, and strategic recommendations that support:

- Underwriting Decisions
- Pricing Strategy
- Financial Planning
- Risk Management
- Preventive Healthcare Initiatives
- Long-Term Business Performance

This approach demonstrates how data analytics can deliver measurable value by transforming raw healthcare data into actionable business intelligence.

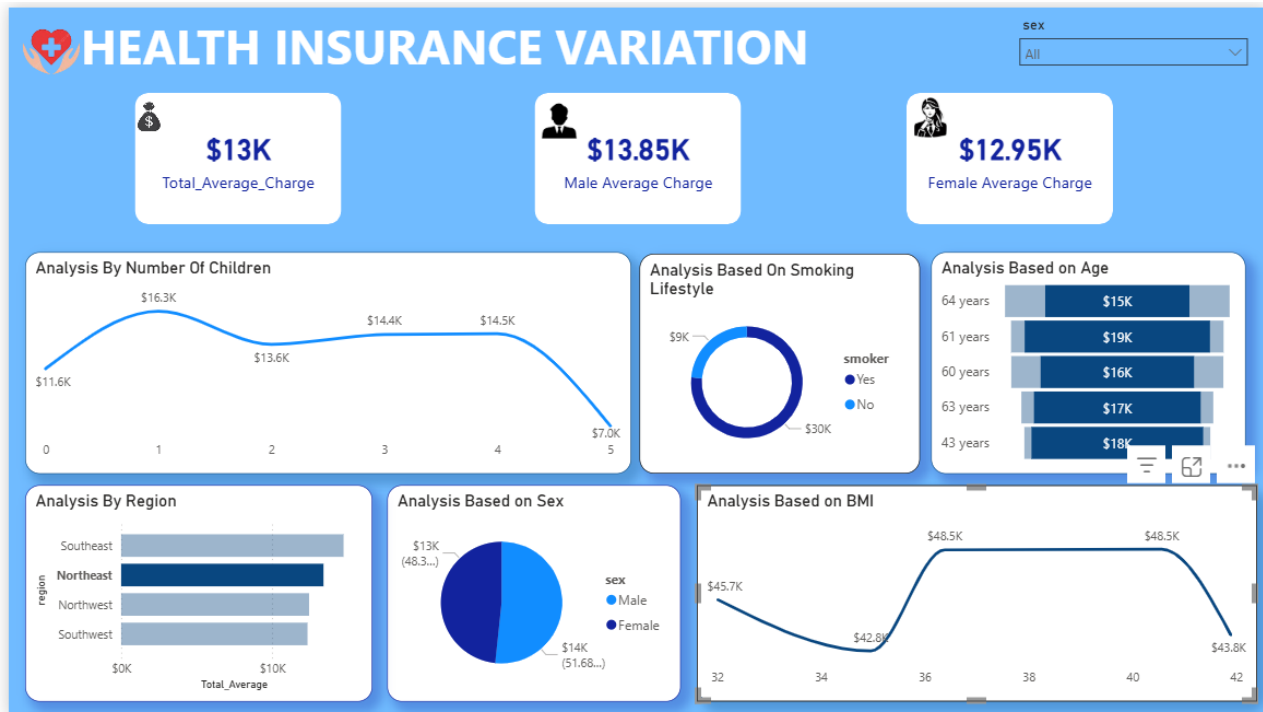
Dashboard Overview

The Healthcare Insurance Cost Analysis dashboard was developed using **Microsoft Power BI** to provide executives, underwriting teams, finance professionals, and business stakeholders with an interactive view of the factors influencing healthcare insurance costs.

The dashboard transforms raw insurance data into meaningful business intelligence through KPI cards, dynamic visualisations, and interactive filtering capabilities that enable users to explore customer behaviour, identify cost drivers, and support evidence-based decision-making.

Rather than presenting static reports, the dashboard provides an analytical environment where decision-makers can quickly assess healthcare cost trends across different customer segments.

The dashboard consists of several integrated components designed to answer key business questions regarding insurance pricing, customer risk, and financial performance.



These include:

- Overall Average Insurance Charges
- Average Charges by Gender
- Insurance Charges by Smoking Status
- Insurance Charges by Age
- Insurance Charges by Body Mass Index (BMI)
- Insurance Charges by Number of Children
- Insurance Charges by Geographic Region
- Interactive Gender Filter

Together, these visualisations provide a comprehensive overview of the variables that influence healthcare insurance costs and support strategic underwriting decisions.

KPI Analysis

Key Performance Indicators (KPIs) provide executives with a concise summary of organisational performance while enabling rapid identification of significant business trends.

The dashboard presents three primary executive KPIs.

Overall Average Insurance Charge

Average Insurance Charge: \$13,000

This KPI represents the average healthcare insurance cost across all policyholders included in the dataset.

It establishes a benchmark against which customer segments can be compared and provides management with an overall view of portfolio pricing performance.

Average Male Insurance Charges

Average Male Charge: \$13,850

Male policyholders demonstrate slightly higher average insurance charges compared to female customers.

However, the difference is relatively small, suggesting that gender alone is not a significant determinant of healthcare expenditure.

This finding indicates that pricing models should place greater emphasis on behavioural and health-related variables rather than demographic characteristics alone.

Average Female Insurance Charges

Average Female Charge: \$12,950

Female policyholders exhibit average insurance costs comparable to those of male customers.

The relatively small variation reinforces the conclusion that gender contributes minimally to overall healthcare expenditure once additional variables are considered.

This supports the development of fairer, evidence-based pricing strategies focused on measurable health risks rather than demographic assumptions.

Executive Insights

The dashboard identifies several important business insights that can improve underwriting decisions, pricing strategies, financial planning, and long-term profitability.

Executive Insight 1 — Smoking is the Primary Driver of Healthcare Costs

Among all variables analysed, smoking status has the strongest influence on insurance charges.

The dashboard demonstrates that smokers incur healthcare costs approximately three times higher than non-smokers.

This substantial difference reflects the increased likelihood of chronic illnesses associated with smoking, including cardiovascular disease, respiratory disorders, stroke, and various forms of cancer.

Higher medical utilisation leads directly to:

- Increased claim frequency
- Larger average claim values
- Higher financial exposure
- Increased underwriting risk

For insurance providers, smoking status represents one of the most valuable predictors of future healthcare expenditure and should remain a central component of premium pricing models.

Executive Insight 2 — Healthcare Costs Increase with Age

The analysis demonstrates a clear positive relationship between age and healthcare expenditure.

As policyholders become older, insurance charges increase steadily due to greater healthcare utilisation and a higher prevalence of chronic medical conditions.

Older customers typically require:

- More frequent medical consultations
- Increased prescription medication
- Additional specialist treatments
- Higher rates of hospital admission

These factors contribute to higher insurance claims and justify gradual increases in premium pricing based on age-related risk.

Understanding this relationship enables insurers to improve actuarial forecasting while maintaining sustainable pricing structures.

Executive Insight 3 — Body Mass Index (BMI) is a Significant Cost Predictor

The dashboard highlights a strong relationship between Body Mass Index (BMI) and healthcare insurance costs.

Customers with higher BMI values consistently incur greater medical expenses than individuals within healthier weight ranges.

Elevated BMI is associated with increased incidence of:

- Diabetes
- Hypertension
- Cardiovascular disease
- Kidney disease
- Musculoskeletal disorders

The financial impact becomes even greater when obesity is combined with smoking, creating one of the highest-risk customer segments within the portfolio.

These findings reinforce the importance of incorporating health indicators into insurance pricing strategies.

Executive Insight 4 — Family Size Moderately Influences Insurance Costs

The analysis indicates that insurance charges increase gradually as the number of dependants increases.

Families generally require greater access to healthcare services, including:

- Child healthcare
- Vaccinations
- Maternity care
- Routine medical consultations

Although family size contributes to higher healthcare utilisation, its overall impact is considerably smaller than lifestyle factors such as smoking and BMI.

This suggests that insurers should assign proportionally lower weighting to the number of dependants within pricing models.

Executive Insight 5 — Regional Differences are Relatively Small

The Southeast region records slightly higher average insurance charges than other geographic regions.

However, the variation across regions is relatively modest compared with behavioural and health-related variables.

This finding suggests that customer location alone provides limited predictive value for future healthcare expenditure.

Consequently, pricing strategies should prioritise health characteristics over geographic factors when assessing insurance risk. :contentReference[oaicite:6]{index=6}

Executive Insight 6 — Gender has Minimal Predictive Value

The dashboard reveals only minor differences between average healthcare charges for male and female customers.

Once variables such as:

- Smoking Status
- Body Mass Index
- Age

are considered, gender contributes very little additional explanatory value.

This demonstrates that behavioural and clinical risk factors are significantly more influential than demographic characteristics in determining healthcare costs.

Adopting this evidence-based approach supports more equitable and transparent underwriting practices.

Executive Insight 7 — Lifestyle Factors Drive Long-Term Financial Risk

The combined analysis demonstrates that lifestyle behaviours are substantially stronger predictors of insurance expenditure than static demographic variables.

Customers who smoke and maintain high BMI levels consistently generate the greatest healthcare costs, making them the highest financial risk group within the insurance portfolio.

These findings support investment in:

- Preventive healthcare programmes
- Wellness initiatives
- Smoking cessation programmes
- Weight management interventions

Reducing lifestyle-related health risks has the potential to lower future claims, improve customer health outcomes, and strengthen long-term profitability.

Executive Insight 8 — Business Intelligence Enables Better Strategic Decisions

This project illustrates how interactive dashboards transform operational healthcare data into actionable business intelligence.

Rather than relying solely on historical reporting, executives can use the dashboard to:

- Identify emerging healthcare cost trends
- Evaluate customer risk profiles
- Optimise underwriting policies
- Improve premium pricing strategies
- Support financial forecasting
- Enhance operational performance

By integrating SQL data extraction with Power BI visualisation, the project demonstrates how modern analytics supports faster, more informed, and evidence-based decision-making across healthcare insurance organisations.

Business Impact

Beyond developing an interactive dashboard, this project demonstrates how data analytics can generate measurable business value across underwriting, financial planning, operational efficiency, customer management, and strategic decision-making.

The analytical findings enable healthcare insurers to move beyond reactive reporting towards proactive, evidence-based management of healthcare costs and insurance risk.

Improved Underwriting Decisions

The analysis identifies the variables that most accurately predict future healthcare expenditure, enabling insurers to refine underwriting policies based on objective evidence rather than assumptions.

By incorporating factors such as:

- Smoking Status
- Body Mass Index (BMI)

- Age
- Family Size

underwriters can make more accurate pricing decisions that better reflect individual risk profiles.

Improved underwriting reduces uncertainty while supporting fair and sustainable premium pricing.

Enhanced Risk-Based Pricing

One of the most significant business outcomes of this project is the ability to support risk-based pricing strategies.

The analysis demonstrates that behavioural and health-related characteristics have considerably greater predictive power than demographic variables such as gender or geographic location.

This enables insurance companies to:

- Price policies more accurately.
- Improve fairness in premium calculation.
- Reduce adverse selection.
- Better align premiums with expected healthcare costs.

More accurate pricing contributes directly to improved financial performance.

Reduced Claims Costs

Identifying high-risk customer groups before policy issuance allows insurers to implement proactive risk management strategies.

Customers exhibiting multiple risk factors—particularly smoking combined with high BMI—represent significantly greater financial exposure.

Early identification enables organisations to:

- Promote preventive healthcare initiatives.
- Encourage healthier behaviours.
- Reduce future claim frequency.
- Lower average claim values.

Reducing healthcare claims contributes directly to improved underwriting profitability.

Improved Customer Segmentation

The dashboard enables management to classify customers according to their healthcare risk characteristics.

Rather than treating policyholders as a homogeneous population, insurers can develop targeted strategies for different customer groups.

Examples include:

- Wellness programmes for high-risk customers.
- Premium incentives for healthy lifestyles.
- Preventive healthcare campaigns.
- Personalised customer engagement.

Customer segmentation strengthens both operational effectiveness and customer satisfaction.

Greater Operational Efficiency

Manual analysis of healthcare insurance data can be time-consuming and resource-intensive.

The combination of SQL data extraction and Power BI reporting automates much of the analytical process, allowing decision-makers to access real-time insights through interactive dashboards.

Automation reduces manual reporting effort while improving consistency, accuracy, and reporting speed.

Stronger Executive Decision-Making

Executives require concise, evidence-based information when making strategic decisions.

The dashboard developed in this project provides management with a clear understanding of:

- Cost drivers.
- Customer risk profiles.
- Pricing performance.
- Healthcare expenditure trends.

This supports more informed decision-making across underwriting, finance, operations, and corporate strategy.

Long-Term Business Sustainability

The project demonstrates how business intelligence contributes to sustainable organisational performance.

By improving pricing accuracy, reducing claims costs, and supporting preventive healthcare initiatives, insurers can achieve:

- Higher profitability.
- Improved customer retention.
- Better financial resilience.
- Stronger competitive advantage.

Data-driven decision-making therefore becomes a strategic capability rather than simply a reporting function.

Financial & Accounting Perspective

Healthcare insurance organisations operate within a complex financial environment where underwriting decisions directly influence profitability, cash flow, reserves, and shareholder value.

The analytical findings generated in this project have important implications across both financial reporting and long-term business performance.

Revenue Generation

Insurance premiums represent the primary source of revenue for healthcare insurers.

Higher-risk policyholders generally pay higher premiums because they are expected to generate greater future healthcare costs.

However, premium pricing must accurately reflect underlying risk.

If premiums are underestimated, future claims may exceed revenue collected, placing pressure on financial performance.

Conversely, excessively high premiums may reduce competitiveness and customer retention.

The analysis therefore supports balanced pricing strategies that optimise both profitability and market competitiveness. :contentReference[oaicite:0]{index=0}

Claims Expense Management

Healthcare claims represent one of the largest operating expenses for insurance companies.

The analysis demonstrates that smoking and obesity significantly increase expected medical costs.

As claim frequency and claim severity increase:

- Operating profit declines.
- Underwriting margins decrease.
- Net income falls.

Understanding these relationships enables insurers to implement targeted interventions that reduce future claims while strengthening financial performance. :contentReference[oaicite:1]{index=1}

Profitability

The project highlights the direct relationship between customer health profiles and organisational profitability.

Healthy policyholders typically:

- Generate fewer medical claims.
- Require less intensive healthcare services.
- Contribute positively to underwriting margins.

Maintaining a balanced portfolio containing a greater proportion of lower-risk customers supports sustainable profitability while reducing financial volatility.

Cash Flow Management

Insurance companies collect premiums before claims are paid.

Effective risk assessment improves the predictability of future claim payments, allowing management to maintain healthier cash reserves.

Conversely, unexpectedly high claim volumes may reduce available liquidity and restrict future operational flexibility.

Accurate analytics therefore supports stronger cash flow forecasting and treasury management.

Insurance Liabilities

Insurance providers maintain claim reserves to cover future obligations.

Higher-risk customer segments require larger reserves due to increased expected claim activity.

As portfolio risk increases:

- Insurance liabilities increase.
- Capital requirements may rise.
- Financial flexibility may decline.

Improved customer risk assessment helps optimise reserve calculations while strengthening financial stability.

Retained Earnings and Shareholder Value

Excessive healthcare claims reduce profitability, which subsequently affects retained earnings and shareholders' equity.

Improving underwriting accuracy enables insurers to:

- Protect net income.
- Strengthen retained earnings.
- Preserve shareholder value.
- Support sustainable long-term growth.

Analytics therefore contributes directly to both operational performance and corporate financial health.

Investment Performance

Insurance companies invest premium income until claims become payable.

Lower claims costs allow a greater proportion of premium income to remain invested, generating additional investment returns.

Consequently, improvements in underwriting quality create financial benefits extending beyond insurance operations into investment performance and overall organisational profitability.

Strategic Financial Decision-Making

The analytical framework developed in this project supports executive decision-making across multiple financial functions, including:

- Premium Pricing
- Underwriting Strategy
- Claims Management
- Financial Planning
- Risk Management
- Capital Allocation
- Long-Term Profitability

By integrating SQL analytics with financial interpretation, the project demonstrates how modern Data Analysts contribute not only to reporting but also to strategic financial management.

Business Intelligence Perspective

This project illustrates the growing role of Business Intelligence within healthcare insurance organisations.

Rather than using dashboards solely for visualisation, the solution enables executives to transform operational healthcare data into actionable insights that support organisational strategy.

The combination of SQL and Power BI enables decision-makers to:

- Identify healthcare cost trends.
- Understand customer behaviour.
- Improve pricing models.
- Optimise underwriting decisions.
- Monitor portfolio performance.
- Evaluate financial risk.
- Support executive reporting.

This demonstrates how Business Intelligence serves as a bridge between operational data, financial performance, and strategic leadership.

Executive Insights

The analysis generated several strategic findings that can improve organisational performance.

Executive Insight 1 — Lifestyle Behaviours Drive Insurance Costs

Smoking and obesity are substantially stronger predictors of healthcare expenditure than demographic characteristics.

These variables should receive greater weighting within pricing models.

Executive Insight 2 — Preventive Healthcare Creates Financial Value

Investing in wellness programmes and preventive healthcare initiatives can reduce future medical claims while improving customer health outcomes.

Preventive care represents both a healthcare strategy and a financial strategy.

Executive Insight 3 — Risk-Based Pricing Improves Sustainability

Pricing policies aligned with measurable healthcare risk produce more equitable premiums while protecting long-term profitability.

Executive Insight 4 — Data Analytics Supports Better Executive Decisions

Interactive dashboards enable executives to evaluate organisational performance quickly while improving the quality of strategic decision-making.

Executive Insight 5 — Business Intelligence Creates Competitive Advantage

Insurance organisations that integrate advanced analytics into underwriting and pricing processes are better positioned to improve profitability, reduce financial risk, and deliver greater customer value.

Strategic Recommendations

The findings from this analysis demonstrate that healthcare insurance organisations can significantly improve underwriting accuracy, financial performance, and customer outcomes by embedding data analytics into their decision-making processes.

Rather than relying primarily on traditional actuarial assumptions, insurers should leverage SQL, Business Intelligence, and predictive analytics to identify cost drivers, optimise pricing strategies, and proactively manage healthcare risks.

Based on the analytical findings, the following strategic recommendations are proposed.

1. Adopt Risk-Based Premium Pricing

Insurance pricing should prioritise measurable health and behavioural indicators rather than demographic characteristics alone.

Premium models should place greater emphasis on:

- Smoking Status
- Body Mass Index (BMI)
- Age

- Medical Risk Factors
- Previous Healthcare Utilisation

This approach enables insurers to establish more equitable pricing structures while improving underwriting profitability.

2. Invest in Preventive Healthcare Programmes

The analysis demonstrates that smoking and obesity are the strongest drivers of healthcare expenditure.

Insurance providers should invest in preventive healthcare initiatives including:

- Smoking cessation programmes
- Weight management support
- Wellness education
- Preventive health screenings
- Lifestyle coaching

Encouraging healthier behaviours reduces long-term claims costs while improving customer wellbeing.

3. Reward Low-Risk Policyholders

Customer retention can be strengthened by recognising healthy behaviours.

Insurance organisations should introduce incentives such as:

- Premium discounts
- Wellness rewards
- Reduced deductibles
- Loyalty programmes
- Preventive healthcare benefits

Rewarding healthier customers encourages long-term engagement while improving portfolio quality.

4. Expand Predictive Analytics Capabilities

Although this project focuses on descriptive analytics, future enhancements should incorporate predictive modelling techniques capable of forecasting future healthcare costs.

Potential analytical models include:

- Logistic Regression
- Random Forest
- Gradient Boosting
- Decision Trees
- Machine Learning Classification Models

Predictive analytics enables insurers to identify high-risk policyholders before expensive claims occur, supporting proactive intervention strategies.

5. Strengthen Executive Reporting

Interactive Business Intelligence dashboards should become an integral component of executive reporting.

Management should continuously monitor:

- Average Insurance Charges
- Customer Risk Profiles
- Premium Performance
- Claims Trends
- Healthcare Cost Drivers
- Portfolio Profitability

Real-time reporting enables executives to respond more rapidly to emerging risks while improving organisational agility.

6. Improve Customer Segmentation

The project demonstrates that policyholders exhibit distinct healthcare risk profiles.

Insurance providers should use analytical segmentation to develop targeted strategies for:

- High-risk customers
- Low-risk customers
- Families
- Older adults
- Wellness-focused customers

Segmentation supports personalised customer engagement while improving underwriting performance.

7. Integrate Business Intelligence Across Departments

Healthcare analytics should not operate in isolation.

The Business Intelligence framework developed in this project can support collaboration across:

- Underwriting
- Finance
- Claims Management
- Customer Experience
- Executive Leadership
- Risk Management

Integrated decision-making improves organisational efficiency while ensuring consistent strategic priorities.

8. Build an Enterprise Healthcare Analytics Platform

The analytical solution presented in this project provides an excellent foundation for enterprise-wide healthcare analytics.

Future enhancements could integrate:

- Electronic Health Records
- Claims Systems
- Customer Relationship Management (CRM)
- Financial Systems
- Fraud Detection Platforms
- Predictive Risk Models

Developing a unified analytics platform would enable healthcare insurers to make faster, more informed decisions while improving long-term financial sustainability.

Skills Demonstrated

This project demonstrates a comprehensive combination of technical, analytical, financial, and business competencies expected of Data Analysts, Business Intelligence Analysts, Healthcare Analysts, Financial Analysts, Commercial Analysts, and Business Analysts working within multinational organisations.

SQL & Database Development

- Microsoft SQL Server
- SQL Query Development
- Data Extraction
- Filtering
- Sorting
- Aggregate Functions
- Data Preparation
- Relational Data Analysis

Microsoft Power BI

- Interactive Dashboard Development
- KPI Card Design
- DAX Measures
- Power Query
- Data Modelling
- Interactive Filters
- Slicers
- Executive Dashboard Design

Healthcare Analytics

- Healthcare Cost Analysis
- Insurance Pricing Analysis

- Customer Risk Assessment
- Healthcare Expenditure Analysis
- Lifestyle Risk Evaluation
- Preventive Healthcare Analytics
- Customer Segmentation

Financial Analytics

- Premium Pricing Analysis
- Claims Cost Assessment
- Revenue Analysis
- Profitability Evaluation
- Cash Flow Interpretation
- Reserve Analysis
- Financial Performance Assessment

Business Intelligence

- Data Visualisation
- Executive Reporting
- KPI Development
- Business Performance Analysis
- Strategic Decision Support
- Business Storytelling

Business Analysis

- Problem Identification
- Business Requirement Interpretation
- Strategic Thinking
- Stakeholder Communication
- Decision Support
- Analytical Reporting

Data Storytelling

This project demonstrates the ability to transform complex healthcare datasets into executive-level business insights.

Rather than presenting technical outputs alone, the analysis explains how healthcare cost drivers influence pricing strategies, financial performance, customer management, and organisational decision-making.

This capability is highly valued within multinational organisations where analysts must communicate technical findings to non-technical stakeholders.

Why This Project Matters

Healthcare insurance organisations generate vast quantities of operational and financial data every day.

Without effective analytical tools, identifying the factors driving healthcare expenditure becomes increasingly difficult, limiting the organisation's ability to price policies accurately, manage financial risk, and improve operational performance.

This project demonstrates how SQL and Microsoft Power BI can transform raw insurance data into actionable business intelligence that supports better decision-making.

By analysing demographic characteristics alongside behavioural and lifestyle factors, the project identifies the variables most strongly associated with healthcare costs and translates these findings into practical business recommendations.

The project illustrates how analytics can help insurers:

- Improve underwriting decisions.
- Optimise premium pricing.
- Reduce future claims costs.
- Strengthen financial planning.
- Support preventive healthcare initiatives.
- Enhance customer engagement.
- Improve organisational profitability.

Beyond technical implementation, the project reflects the evolving role of modern Data Analysts as strategic business partners capable of connecting data, finance, healthcare operations, and executive decision-making.

About the Author

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Dayo Rex Afariogun is a Business Intelligence and Data Analytics professional with over 14 years of experience helping organisations improve business performance through data-driven decision-making, financial analysis, and strategic business intelligence.

With an academic background in **Accounting** and hands-on expertise in **SQL, Python, Microsoft Power BI, Microsoft Excel, and Business Intelligence**, he specialises in transforming complex datasets into actionable insights that support executive leadership and organisational growth.

His portfolio spans multiple analytical domains, including:

- Healthcare Analytics
- Financial Analytics
- Commercial Revenue Analysis
- Credit Risk Analytics
- Banking Analytics
- Business Intelligence
- Executive Dashboard Development
- Operational Performance Analysis

His projects demonstrate practical experience in combining technical expertise with business understanding to deliver analytical solutions that improve profitability, optimise operations, strengthen risk management, and support evidence-based decision-making.

Dayo is passionate about leveraging analytics to solve complex business problems and is actively seeking opportunities to contribute his expertise within multinational organisations where innovation, data, and strategy drive sustainable business success.

Connect with me:

- **LinkedIn:** www.linkedin.com/in/dayo-afariogun

- **GitHub:** github.com/dayo-data-analyst

Executive Conclusion

This project demonstrates how healthcare insurance data can be transformed into meaningful business intelligence that supports underwriting decisions, pricing strategy, financial management, and executive decision-making.

Using **Microsoft SQL Server** for data extraction and **Microsoft Power BI** for interactive visualisation, the project analysed the relationship between customer demographics, lifestyle characteristics, and healthcare insurance costs. Through structured data preparation, KPI development, dashboard design, and business analysis, the solution identified the key drivers of healthcare expenditure across the insurance portfolio.

The analysis revealed that lifestyle factors—particularly smoking and elevated Body Mass Index (BMI)—are significantly stronger predictors of insurance costs than demographic variables such as gender or geographic region. Age also demonstrated a meaningful relationship with healthcare expenditure, while family size contributed a moderate influence on insurance charges. These findings provide insurers with valuable insights for developing more accurate risk-based pricing models and strengthening underwriting decisions.

From a financial perspective, the project highlights how data analytics contributes to improved premium pricing, reduced claims costs, enhanced profitability, stronger reserve management, healthier cash flow, and long-term shareholder value. The analytical framework also supports preventive healthcare initiatives that can reduce future medical claims while improving customer outcomes.

From a professional perspective, the project showcases practical expertise in:

- SQL Development
- Microsoft Power BI
- Healthcare Analytics
- Insurance Analytics
- Business Intelligence
- Financial Analysis
- Executive Dashboard Design
- KPI Development
- Data Storytelling
- Strategic Business Analysis

Ultimately, this project illustrates how modern Business Intelligence enables organisations to move beyond historical reporting towards proactive, data-driven decision-making. It reflects the role of today's Data Analyst as a strategic partner capable of transforming operational healthcare data into insights that improve financial performance, optimise customer outcomes, and support long-term organisational success.

Key Technologies

- Microsoft SQL Server
- SQL
- Microsoft Power BI
- Power Query
- DAX
- KPI Cards
- Interactive Dashboards
- Data Modelling
- Business Intelligence

Core Competencies Demonstrated

- Healthcare Analytics
- Insurance Cost Analysis
- Business Intelligence
- SQL Development
- Power BI Dashboard Development
- Financial Analytics
- Customer Segmentation
- Risk-Based Pricing
- Executive Reporting
- Data Storytelling
- Strategic Business Analysis
- Decision Support

Thank you for taking the time to review this project.

I welcome opportunities to discuss how my experience in **Business Intelligence, SQL development, Power BI, healthcare analytics, financial analysis, and strategic data storytelling** can help organisations transform complex data into actionable insights that improve business performance and support evidence-based decision-making.